Jianman Lin

Embodied Intelligence · Multimodal Learning · Generative Modeling

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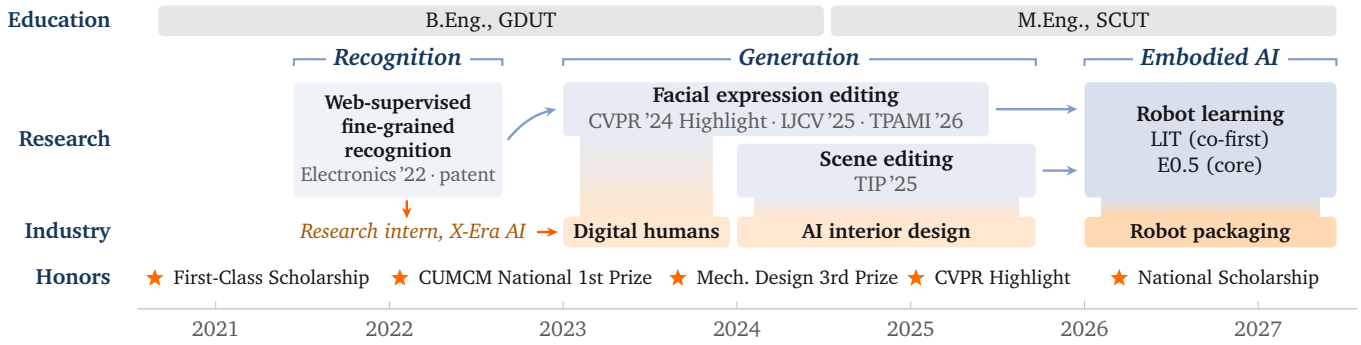
Research Profile

In my sophomore year, I joined **Prof. Tianshui Chen**'s group and began research on **web-supervised fine-grained image recognition**. I then moved to **generative models** for **facial expression editing** and **scene editing**, with industrial applications in digital humans and interior design. Deploying VLA policies on robots, I observed that they exploit task-irrelevant visual cues, which motivated **LIT**; I also contributed to the 3D world–action model **E0.5**. Publications include first/co-first-author papers in **TPAMI, IJCV, and TIP** and a **CVPR Highlight**.



Education

South China University of Technology (SCUT) — Guangzhou, China
M.Eng. in Electronic Information — Sep. 2024 – June 2027 (expected)
Advisors: Prof. Tianshui Chen and Prof. Chunmei Qing.

Guangdong University of Technology (GDUT) — Guangzhou, China
B.Eng. in Industrial Engineering (GPA 87.2/100) — Sep. 2020 – June 2024

Publications

[1] **Breaking the Vision–Action Shortcut: Latent Interface Training for Generalizable Robot Foundation Models**
Jianman Lin, Shailesh Shailesh, Zhongyi Luo, Jiafei Duan.
arXiv:2609.12641, 2026; under review at **ICRA**. **Co-first author**

[2] **E0.5: From Reconstructive to Future-Predictive 3D Foundations for Robotic World-Action Modeling**
X-Era AI Lab Team (26 authors), with **Jianman Lin** as a core contributor.
arXiv:2610.36011, 2026.

[3] **Learning Spatial-Temporal Coherent Correlations for Speech-Preserving Facial Expression Manipulation**
Tianshui Chen, **Jianman Lin**, Zhijing Yang, Chunmei Qing, Guangrun Wang, Liang Lin.
IEEE TPAMI, 2026. **Co-first author**

[4] **Neural Scene Designer: Self-Styled Semantic Image Manipulation**
Jianman Lin, Tianshui Chen, Chunmei Qing, Zhijing Yang, Shuangping Huang, Yuheng Ren, Liang Lin.
IEEE TIP, 2025. **First author**

[5] **Contrastive Decoupled Representation Learning and Regularization for Speech-Preserving Facial Expression Manipulation**
Tianshui Chen, **Jianman Lin**, Zhijing Yang, Chunmei Qing, Yukai Shi, Liang Lin.
IJCV, 2025. **Co-first author**

[6] **Learning Adaptive Spatial Coherent Correlations for Speech-Preserving Facial Expression Manipulation**
Tianshui Chen, **Jianman Lin**, Zhijing Yang, Chunmei Qing, Liang Lin.
CVPR, 2024. **Highlight (top 2.8%)**.

Research Experience

LIT: Generalizable Robot Foundation Models [1] 2026 – Present

- **Motivation.** Robot policies that perform well on the training distribution often degrade under visual distribution shifts (e.g., lighting or viewpoint changes) by exploiting vision–action shortcuts. *How can action learning and visual conditioning be structured to mitigate these shortcuts while preserving the spatial information needed to act?*
- **Method.** A two-stage strategy: an image-free action prior conditioned on the target end-effector pose, followed by a pose-supervised latent interface that serves as the sole visual pathway to actions.
- **Results.** Matched in-distribution performance; zero-shot LIBERO-Plus success **+3.9–10.7 pp** across four VLA and world–action models; real-robot success **+13.3–16.7 pp** under unseen visual conditions.
- **My role.** Identified the problem, developed the method, ran all simulation experiments, and wrote the manuscript under Prof. Jiafei Duan's guidance; the co-first author ran the real-robot experiments.

Speech-Preserving Facial Expression Manipulation [3, 5, 6] 2023 – 2025

- **Motivation.** Speech-preserving facial expression manipulation (SPFEM) edits the facial expression of a speaker or digital human while keeping the mouth shapes consistent with the speech, yet suitable paired training data are scarce. *How can explicit paired supervision be provided for SPFEM when paired data is scarce?*
- **Method.** Derived supervision from the inputs themselves: emotion-invariant facial correlations as training objectives (ASCCL, STCCL) and contrastively disentangled content and emotion representations (CDRL).
- **Results.** Improved expression accuracy and lip synchronization across editing backbones and benchmarks.
- **My role.** Led all three works from idea to manuscript; my advisor revised the manuscripts.

Style-Consistent Scene Editing [4] 2024 – 2025

- **Motivation.** Text-guided image editing often yields content that is stylistically inconsistent with the surrounding scene. *How can edits follow the user's intent while remaining consistent with the scene's style?*
- **Method.** A diffusion-based editor with parallel text and style cross-attention; style representations learned by contrasting regions within and across images.
- **Results.** Established a benchmark with style metrics; improved text alignment and style consistency.
- **My role.** Led the idea, method, benchmark, experiments, and writing; my advisor revised the manuscript.

Industry Experience

Research Intern: AI & Robotics Dec. 2023 – Present
X-Era AI & ExpanderaAI Lab Guangzhou, China

- **Robotic flexible packaging (2026 – Present), project lead.** Led the design of a dual-arm system that packs items into a carton and seals it, a ∼5-minute long-horizon task achieving **90% success**, by combining subtask reasoning with offline reinforcement learning; built the pipeline from teleoperation to deployment. [Video]
- **AI interior design (2025 – 2026).** Deployed multimodal large models in the Feiliu AI interior design product: VLM-based floor-plan parsing (98% accuracy), LLM-based layout generation, and agent workflows.
- **Digital humans (2023 – 2024).** Model training and deployment; StyleGAN priors for few-shot, identity-consistent avatar generation.

Honors, Patents & Service

- **Honors:** National Scholarship (2026); CVPR 2024 Highlight; First-Class Scholarship; National First Prize, CUMCM; National Third Prize, Mechanical Innovation Design Competition.
- **Patents:** 3 granted invention patents (2 as first inventor). **Reviewer:** ICRA, CVPR, NeurIPS, ECCV.
- **Language:** English – IELTS Overall **7.0**.